Windows Fundamentals — Instructor Speaker Notes

Slide-by-slide notes for the 23-slide deck DS.01.03 - Windows Fundamentals.pptx. For every slide you get: what to **say**, what to **ask** the class (with the expected answers), what to **demo** live, and what to **watch for**. The class is entry-level — recent high-school graduates with no IT background — so the wording avoids jargon and leans on the deck's analogies.

How to read the labels

- **SAY** — suggested wording. Adapt it to your voice; don't read it aloud word for word.
- **ASK** — questions to pose, with the expected answer in italics. Cold-call by name where you can.
- **DEMO** — live actions on the projector. Practice these once before class.
- **WATCH FOR** — misconceptions and pacing traps.

Timings assume a **~70-minute session including the 15-minute lab**. Shorter period? Compress the demos first — never the lab.

The lesson at a glance

Part	Slides	Content	Time
Welcome & course map	1–2	What the course covers, the four exit skills	5 min
Topic 1 · File types	3–7	Extensions, five families, zip vs iso, executable safety	12 min
Topic 2 · Files & folders	8–11	Naming rules, the five moves, Properties	10 min
Topic 3 · Drives & paths	12–16	Drive letters, file systems, paths, safe eject	12 min
Topic 4 · Command line	17–20	CMD, PowerShell, one job three ways	11 min
Hands-on lab	21	Four tasks inside the practice folder	15 min
Recap & wrap-up	22–23	Pocket reference cold-calls, next steps	5 min

Before class — setup checklist

- Demo machine: turn file name extensions ON (**File Explorer → View → Show → File name extensions**) — the whole of Topic 1 depends on it.

- Seed your **Downloads** folder with a mixed handful: a .docx, a .jpg, a .zip, an .exe installer, an .mp3.
- Prepare a demo folder (e.g. `C:\Demo`) and one spare USB stick with safe contents for the eject demo.
- Optional but effective: one innocent `.zip` to extract live, and one `.iso` to mount.
- Students need PCs (or VMs) where they can create folders inside their own practice folder.
- Keep the **student handout** nearby — every slide maps to a section there, and the 20-question Moodle quiz covers all four topics.

> **ANSWER KEY — ON-SLIDE QUICK CHECK**
>
> **Slide 7:** `invoice.pdf.exe` is a **program** — the last extension is the real type. Correct student action: don't run it, stop and ask.

Slide by slide

Slide 1 · Cover — Windows Fundamentals

WELCOME · ~2 MIN

SAY—"Welcome to Windows Fundamentals. You need zero experience for today. By the end of this session you will read any file name and know what it will do, create and manage files and folders properly, find anything on any drive, and type your very first commands. Four topics, one hands-on lab, no prerequisites."

ASK—Show of hands: who has ever opened the command line — even by accident?

> *Expected: Calibrates your pace. Usually a few hands; reassure the rest that nobody is behind.*

Slide 2 · Course map — four stops, one new skill

ROADMAP · ~3 MIN

SAY—Walk the four stops: file types (spot .exe, .zip, .iso and say what each is for), files & folders (manage by rule, not guesswork), drives & paths (find anything fast, eject media safely), and the command line (dir, cd, copy — and a first cmdlet). Point at the exit skills: "That's what you'll actually be able to DO — and the lab at the end tests exactly those four things."

ASK—Warm-up guess: who can tell me what the .zip at the end of a file name means?

> *Expected: Take 2–3 guesses without confirming — slide 6 pays it off.*

Slide 3 · Topic 01 divider — Understanding File Types

FRAMING · ~30 SEC

SAY—"Topic one: file types. Here's the big idea — the label at the end of a file name isn't decoration. Windows reads it before it does anything else."

Slide 4 · The extension is a label Windows actually reads

CORE CONCEPT · ~3 MIN

SAY—Use report.docx: the part you choose is report, the part after the last dot — .docx — is the extension. Windows uses it for two jobs: picking the icon you see, and choosing which app opens the file when you double-click. Extensions are hidden by default; we turn them on. And renaming holiday.zip to holiday.exe does NOT convert it — the inside never changes, you only changed the label.

DEMO—File Explorer → View → Show → File name extensions, then open your seeded Downloads folder and read the names aloud together.

WATCH FOR—The "renaming converts the file" myth is the stickiest misconception of the whole lesson. Kill it here; it returns on slides 7 and 22.

Slide 5 · The usual suspects — five families

REFERENCE FRAME · ~2 MIN

SAY—"Almost every file you will ever touch belongs to one of five families." Skim the table — programs run things, documents you read and edit, media you watch and hear, containers pack things, system files are Windows' own parts. Don't read every extension aloud.

ASK—Rapid classification, shout it out: .mp3? .msi? .pptx? .dll? .iso?

> *Expected: Media · Programs · Documents · System · Containers.*

SAY—Land the system row: ".dll and .sys are Windows' own parts — look, don't touch. Deleting one .dll can break several apps at once."

Slide 6 · .zip packs it small, .iso clones the whole disc

CONTAINERS · ~2 MIN

SAY—Zip = a suitcase: squeezes files smaller for storage and sending — extract before you actually use what's inside (right-click → Extract All). Windows opens .zip by itself; .rar needs an extra tool. Iso = a photocopy of a whole disc: double-click to mount, and Windows pretends it's a real drive with its own letter — this is how Windows install media is shared. Eject it when done. .cab is Microsoft's own cabinet for updates and drivers — you'll see them,

rarely open them.

DEMO—Optional: extract the innocent zip live; mount the iso and show the new drive letter appear in This PC.

Slide 7 · Double-click means "run this"

KEY SLIDE — SAFETY · ~3 MIN

SAY—Opening a photo is safe; opening an .exe or .msi means run this — it changes the computer. Scripts (.bat, .cmd) are plain-text command lists that execute at full speed. Then the classic trick: photo.jpg.exe. With extensions hidden, Windows shows only photo.jpg — the dangerous .exe stays invisible, and attackers email these as "pictures" or "invoices". The rule: only run executables from sources you trust.

ASK—On-slide quick check: a file shows up as invoice.pdf.exe — what is its real type?

> *Expected: A program — the LAST extension is the truth. Follow-up: "So an unexpected 'invoice' lands in your inbox — what do you do?" Stop and ask; never double-click.*

WATCH FOR—This slide is why extensions stay ON for the rest of the course — say that explicitly.

Slide 8 · Topic 02 divider — Files & Folders in Action

FRAMING · ~30 SEC

SAY—"Now that you can read file names, let's do things with files. Create it, name it, move it — and read its ID card."

Slide 9 · Windows has house rules for names

RULES · ~3 MIN

SAY—Nine characters are banned outright — and each one already has a job: backslash separates folders in a path, the colon belongs to drive letters, * and ? are search wildcards, | redirects in commands. Reserved names like CON, PRN, AUX, NUL are historic device names Windows owns. The full path stays under 260 characters. Capital letters don't matter to Windows — but be consistent for humans. Safe habit: letters, numbers, spaces, hyphens, underscores — always legal. Bonus habit: start dates like 2026-01-31 and files sort themselves.

DEMO—Optional crowd-pleaser: try to create a file named con.txt — Windows refuses.

WATCH FOR—Students will try to memorize the nine characters. Push the why instead — the why is what sticks, and it's what the quiz checks.

Slide 10 · The five moves — and why a file sometimes refuses

SKILLS · ~3 MIN

SAY—Walk the five moves with their shortcuts: new folder (Ctrl+Shift+N), copy (Ctrl+C, Ctrl+V — original stays), move (Ctrl+X, Ctrl+V — original relocates), rename (F2), delete (Delete key → Recycle Bin — undoable). Two traps: Shift+Delete skips the Bin — gone for good. And dragging has a twist: same drive moves, different drive copies; Shift forces a move, Ctrl forces a copy. When a file fights back it's one of three things — it's open in an app, the path is too long, or you lack permission. Technician's order: close it → shorten it → check permissions.

DEMO—Live in your demo folder: copy, move, F2-rename, delete, then open the Recycle Bin and restore. Students copy every keystroke later in the lab.

Slide 11 · Properties is the file's ID card

CONCEPT PAIR · ~2 MIN

SAY—Right-click any file → Properties. General tab: facts — type, location, size, dates — plus two flags. Read-only means look, don't edit; Hidden means invisible unless you choose to show hidden items. Security tab: NTFS permissions — Read, Write, Modify, Full control — deciding who may do what. Then the distinction to write on the board: attributes describe the FILE; permissions describe what PEOPLE may do with it. The attrib and icacls commands show the same information in CMD — they'll meet those soon.

WATCH FOR—Attribute-vs-permission confusion is a classic exam trap. Flag it now as something they'll see again.

Slide 12 · Topic 03 divider — Drives, Media & Paths

FRAMING · ~30 SEC

SAY—"Where do files actually live? Disks, USB sticks — and the addresses that find them."

Slide 13 · One disk can wear many letters

STORAGE MODEL · ~2 MIN

SAY—A physical disk is the hardware. It gets split into partitions — the usable sections Windows calls volumes — and each volume gets a drive letter. C: is where Windows and your installed programs live. Plug in a USB stick and it simply takes the next free letter — and letters aren't owned, so the same stick can be E: today and F: tomorrow. The tool for this layer is Disk Management, diskmgmt.msc — you'll use it in later classes.

SAY—Analogy if it helps: one house, several rooms, each room with its own mailbox

number.

Slide 14 · NTFS, FAT32, exFAT — same shelf, different rules

COMPARISON · ~3 MIN

SAY—A file system is the set of rules a drive uses to store files — you choose one when formatting, and formatting erases everything, so check the drive letter twice. NTFS runs the Windows system drive and brings permissions plus self-repair journaling. FAT32 is ancient but universal — with a hard 4 GB per-file limit. exFAT is FAT32 modernized: huge files, works everywhere. Rule of thumb: formatting a USB stick for class or work — pick exFAT.

ASK—"Your friend says their 32 GB stick is broken because a 5 GB video won't copy onto it — even though 29 GB are free. What's actually wrong?"

> *Expected: The stick is FAT32; the 4 GB limit is per file, not per drive. Fix: reformat to exFAT (after copying everything off).*

Slide 15 · Every file has an address — learn to read it

PATHS · ~3 MIN

SAY—Read C:\Users\Ava\Documents\report.docx left to right: drive letter, folders separated by backslashes, then the file itself. Absolute path — the full address, works from anywhere. Relative path — directions from where you stand; it only works if you're already in the right folder, and dot-dot means one folder up. UNC path — two backslashes then a server name — an address on another computer. And the 260-character limit covers the whole path, so deep folder nests cause problems.

ASK—"You're standing in C:\Users\Ava. Is Documents\report.docx absolute or relative — and what's the absolute version?"

> *Expected: Relative — the absolute version is C:\Users\Ava\Documents\report.docx.*

Slide 16 · Eject first. Your files will thank you.

SAFETY · ~3 MIN

SAY—Windows doesn't always write to a USB stick the instant you save — it holds data in a cache to work faster. Pull the stick mid-write and the file ends up half-written and corrupted. The safe routine: click Safely Remove Hardware, choose the device, wait for the message — then pull. "Device is in use" means something still has a file open: close it and eject again, don't yank. And never plug in a USB drive you found — attackers leave infected sticks where people will pick them up. Found one? Hand it to IT.

DEMO—With the spare stick: copy a file onto it, eject via Safely Remove Hardware, wait for

the message, pull. Ten seconds that prevent the most common data-loss habit.

Slide 17 · Topic 04 divider — The Command Line

FRAMING · ~30 SEC

SAY—"Everything so far was mouse work. Now the same jobs with no clicking — this is where you start thinking like an administrator."

Slide 18 · CMD — a few words do the work

LIVE TYPING · ~4 MIN

SAY—Open it: Win+R, type cmd, Enter. The prompt — C:\Users\Ava> — shows where you stand; everything you type happens in that folder. Type a command, press Enter, read what it says back. The starter vocabulary: dir lists, cd changes folder (cd .. goes up), md makes a folder, rd removes an empty one, copy duplicates, move relocates, del deletes, ren renames, cls clears — and command /? is built-in help for anything.

DEMO—Type the six-line session from the slide exactly, narrating each line. Finish with dir /? so they see the built-in manual.

WATCH FOR—First contact with a blinking cursor intimidates. Repeat the safety net: dir and /? can never hurt anything. And the one real danger: del skips the Recycle Bin — deleted means deleted.

Slide 19 · PowerShell — same idea, superpowers

PATTERN · ~3 MIN

SAY—PowerShell does the same jobs, but its cmdlets read like sentences: Verb-Noun. Get-ChildItem literally means get the child items — that's dir. Learn the pattern once and you can guess commands forever. CMD passes text between commands; PowerShell passes objects — real structured data — so cmdlets chain into pipelines that automate whole jobs. Get-Help explains any cmdlet. And your old friends work here too: dir, cd, copy are aliases.
Scripts: .bat for CMD, .ps1 for PowerShell.

WATCH FOR—Don't deep-dive pipelines today — plant the seed, keep momentum. Pipelines get their own class later.

Slide 20 · One job, three ways

PAYOFF · ~3 MIN

SAY—One job: create a folder called Invoices and move Q1.xlsx into it. Mouse: three steps.

CMD: two lines — md Invoices, move Q1.xlsx Invoices\ — done in five seconds. PowerShell: New-Item, Move-Item — same result, every time. The punchline: for one file, the mouse is fine. For a hundred files, or the same job every Monday — typing wins.

DEMO—Run all three ways live. Time the CMD version on your watch — five seconds lands the point better than any explanation.

Slide 21 · Your turn — the 15-minute lab

LAB · 15 MIN

SAY—"Your turn. Work inside your own practice folder — nothing in this lab can break Windows." Read the four tasks and their 'done when' checks aloud: extensions detective in Downloads; build Projects\2026\Jan with the mouse; repeat with no mouse in CMD and verify in PowerShell; copy Jan to a USB stick and eject safely.

WATCH FOR—Circulate constantly. Task 3 is where most students stall — first real CMD session. Pair fast finishers with strugglers. Push /? and Get-Help before hands go up: technicians read manuals. The 'done when' checks ARE your assessment — initial each one as you pass.

Slide 22 · Pocket reference

RECAP · ~2 MIN

SAY—Rapid-fire recap — cold-call rows by name: "Banned characters?" "Who takes the next free letter?" "Help command in CMD? In PowerShell?" "What do you do with a found USB stick?"

SAY—Tell students to photograph this slide — it's the whole lesson on one page, and the handout's pocket reference mirrors it.

Slide 23 · Wrap-up — you can now…

CLOSE · ~3 MIN

SAY—Read the four "you can now" statements as a checklist and have students nod along: read any file name; create, name, move and inspect files and folders; find anything on any drive and eject safely; run first CMD and PowerShell commands. Then the road ahead: Microsoft Learn and Professor Messer are free, and these topics map directly to CompTIA A+ Core 2. Next class: hands-on lab — user folders, permissions, and your first script.

ASK—Exit ticket, one line each: write one thing you can now do that you couldn't do this morning.

> *Expected: Collect at the door — instant feedback on whether the exit skills landed.*

AFTER CLASS

Assign the student handout for revision (its Check Yourself questions have answers in the back), then run the 20-question Moodle quiz next session — it covers all four topics in nine question types.